

AZ-900T00

Learning path 04: Apply Azure Skills in Guided Projects



Notice and Intent

This guided project set is provided to provide OPTIONS for practical activities during class. It is NOT expected that you complete all guided projects during a delivery. Given the breadth of content, the goal should be to complete 1, or maybe 2 if timing allows, of the guided projects during a delivery.

The students can complete the additional guided projects using a Bring Your Own Subscription model from Microsoft Learn at any time.

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Guided project: Deploy a static website with Azure Blob Storage



Learning Objectives

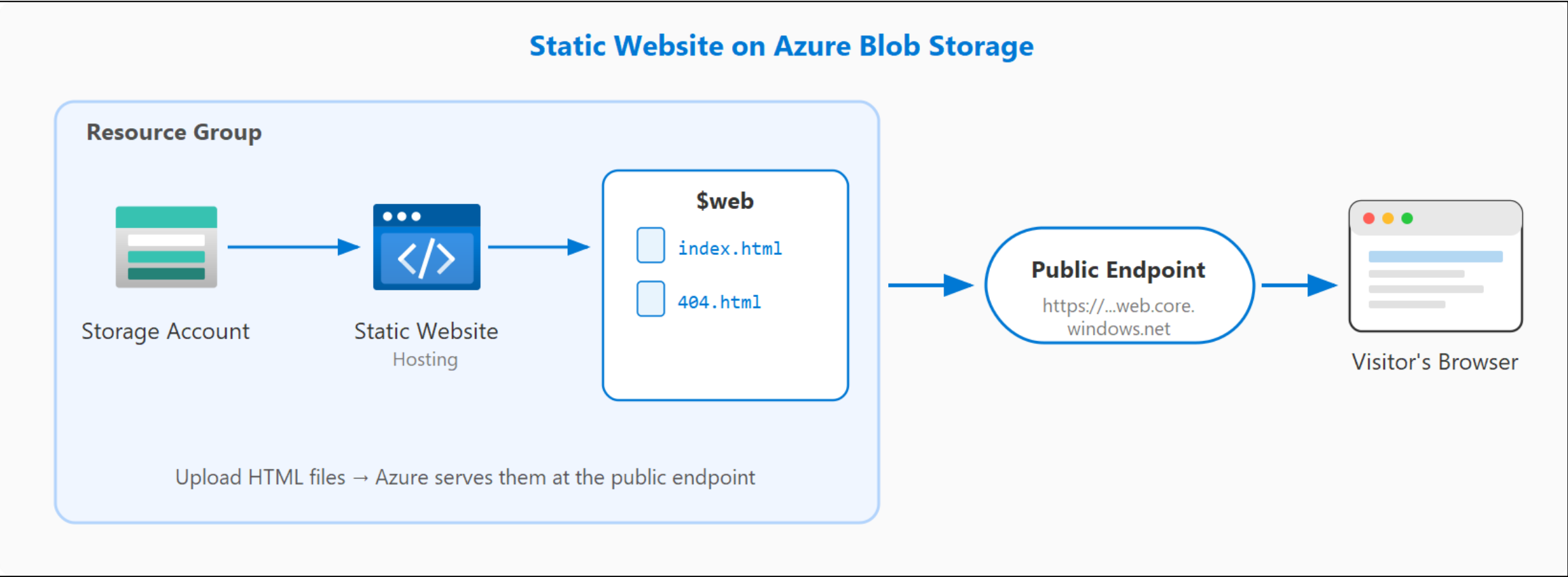
After completing this module, you will be able to:

- 1** Create a storage account and enable static website hosting
- 2** Upload HTML content and verify the public endpoint
- 3** Update site content and confirm the change is live

Project: Static website on Blob Storage objectives

- Scenario: Publish a product landing page without provisioning VMs or web servers
- Enable static website hosting on a storage account to create a public endpoint
- Azure automatically creates a \$web container to serve HTML, CSS, and image files
- Exercise 1: Create a storage account and enable static website hosting
- Exercise 2: Upload HTML content and verify the live site
- Exercise 3: Update the site content and confirm the change

Project: Static website on Blob Storage diagram



Knowledge check



1 What must you enable on a storage account before it can serve static website content?

- ☐ Blob versioning
- ☐ Static website hosting
- ☐ Azure CDN integration

2 Where does Azure store the files for a static website hosted in Blob Storage?

- ☐ A container named blob-website
- ☐ The root of the storage account
- ☐ A container named \$web

3 What happens when you upload an updated file to the \$web container?

- ☐ You must redeploy the storage account
- ☐ The updated content is available without redeployment
- ☐ The previous version enters a staging state

Guided project: Organize and protect resources with tags and locks



Learning Objectives

After completing this module, you will be able to:

1 Create resources and apply tags for organization and cost tracking

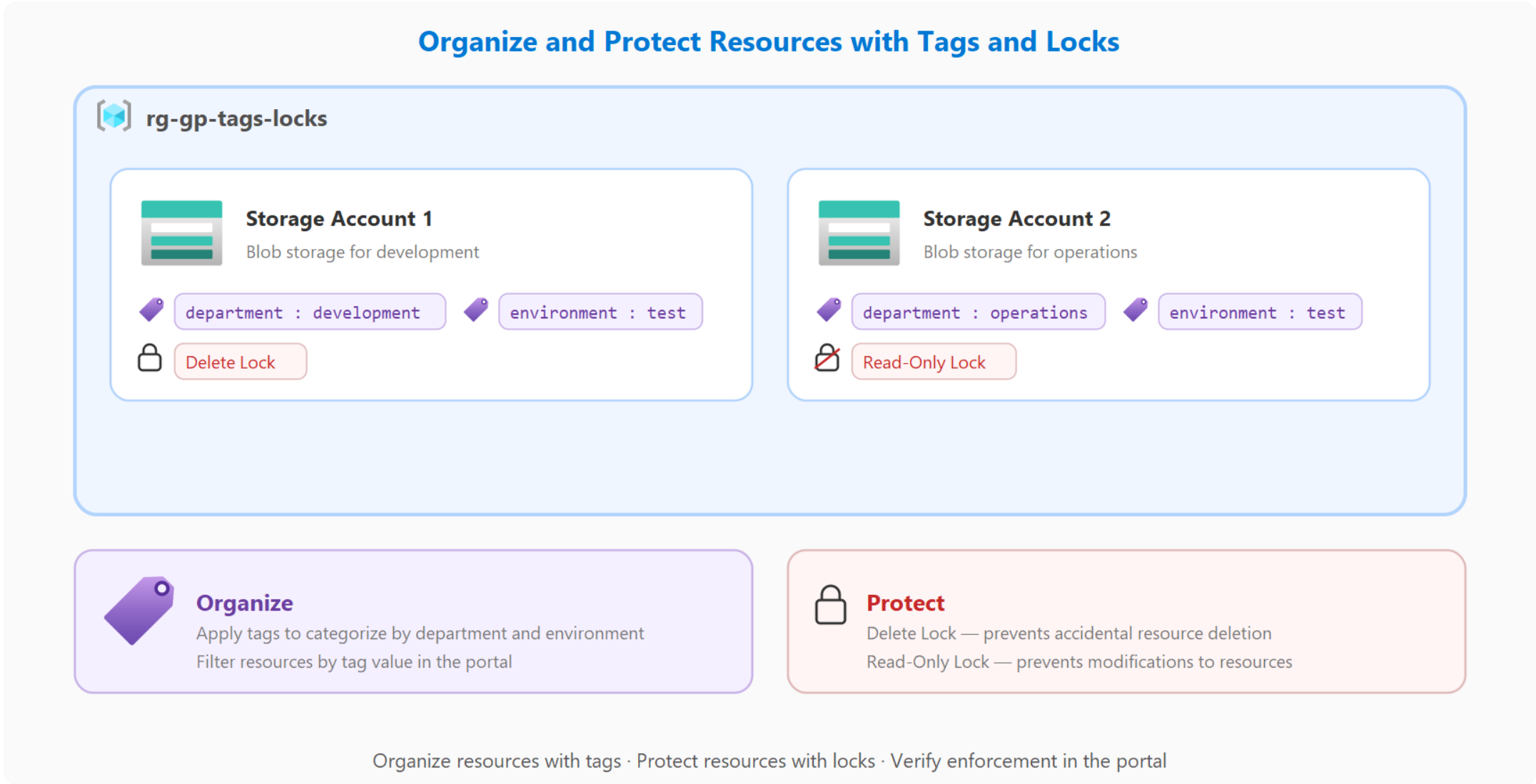
2 Apply a delete lock to prevent accidental resource removal

3 Test lock enforcement and then remove the lock

Project: Tags and locks for governance objectives

- Scenario: Set up a shared Azure environment with organized and protected resources
- Tags are key-value pairs for cost reporting, ownership tracking, and governance
- Locks prevent accidental modification or deletion, even for users with full permissions
- Exercise 1: Create resources and apply organizational tags
- Exercise 2: Apply resource locks to prevent accidental changes
- Exercise 3: Test lock enforcement and confirm the full lock lifecycle

Project: Tags and locks for governance diagram



Knowledge check



- 1** What is the primary purpose of applying resource tags in Azure?
 - ☐ To restrict who can access a resource
 - ☐ To organize resources for cost tracking and management
 - ☐ To prevent resources from being deleted

- 2** What does a read-only resource lock prevent?
 - ☐ Only deletion of the resource
 - ☐ Modifications and deletion of the resource
 - ☐ All access to the resource including viewing

- 3** What must you do before you can delete a resource that has a delete lock?
 - ☐ Disable the lock using Azure Policy
 - ☐ Contact Azure Support to override the lock
 - ☐ Remove the lock first

Guided project: Build a simple website endpoint with Azure Functions



Learning Objectives

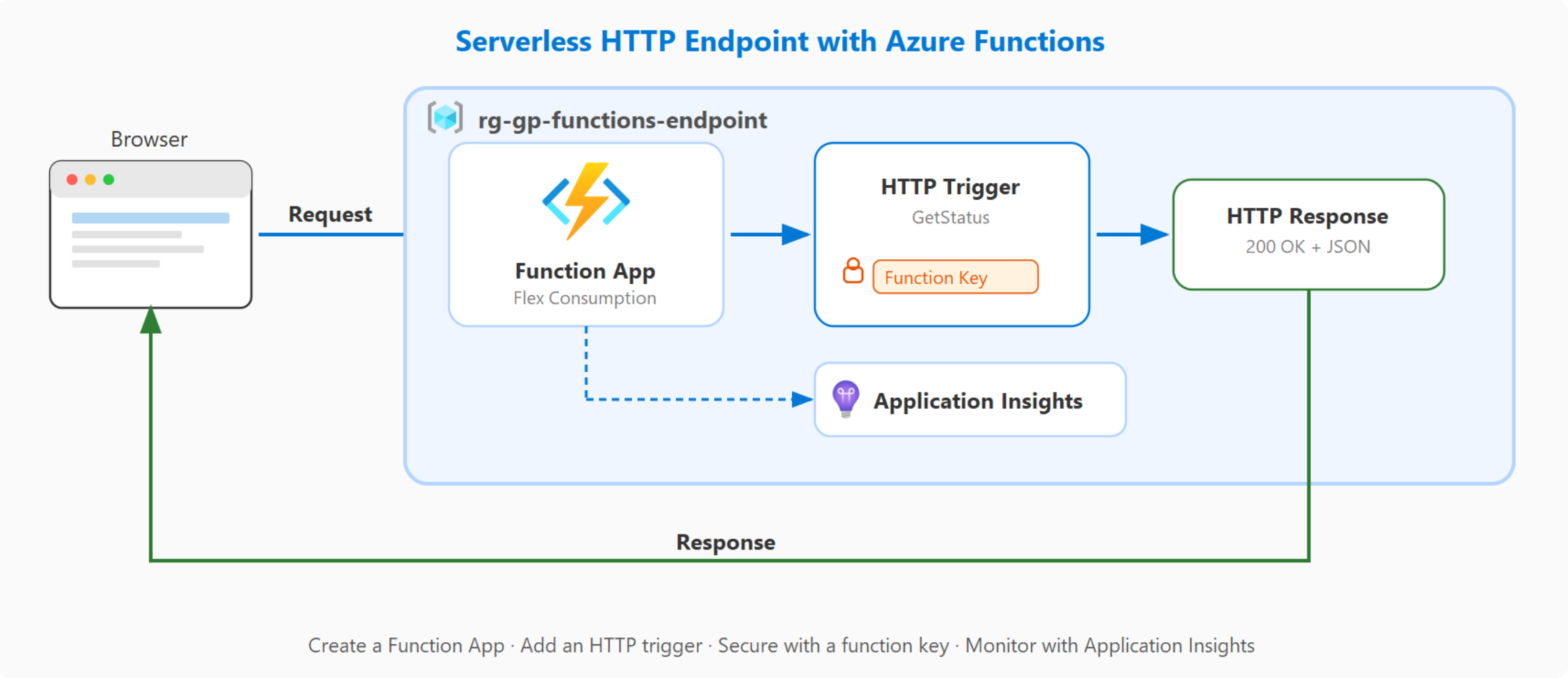
After completing this module, you will be able to:

- 1** Create a Function App in a Flex Consumption plan
- 2** Deploy an HTTP-triggered function from Cloud Shell
- 3** Test endpoint behavior, add security, and review logs

Project: Serverless endpoint with Azure Functions objectives

- Scenario: Create a lightweight contact page endpoint without maintaining servers
- Azure Functions is a serverless compute service that runs code in response to events
- An HTTP trigger gives your function its own URL callable from any browser
- Exercise 1: Create a Function App with the Flex Consumption plan
- Exercise 2: Deploy an HTTP-trigger function from Cloud Shell
- Exercise 3: Test the endpoint, add security, and review logs

Project: Serverless endpoint with Azure Functions diagram



Knowledge check



- 1** What type of trigger allows an Azure Function to respond to web requests?
 - ☐ Timer trigger
 - ☐ HTTP trigger
 - ☐ Blob trigger

- 2** What is the main advantage of using Azure Functions for a simple endpoint?
 - ☐ It provides a dedicated virtual machine
 - ☐ You do not need to manage any infrastructure
 - ☐ It automatically creates a custom domain

- 3** What Azure service lets you monitor function invocations and view execution logs?
 - ☐ Azure Monitor metrics
 - ☐ Azure Activity Log
 - ☐ Application Insights

Guided project: Set up new employee access (Entra ID and RBAC)



Learning Objectives

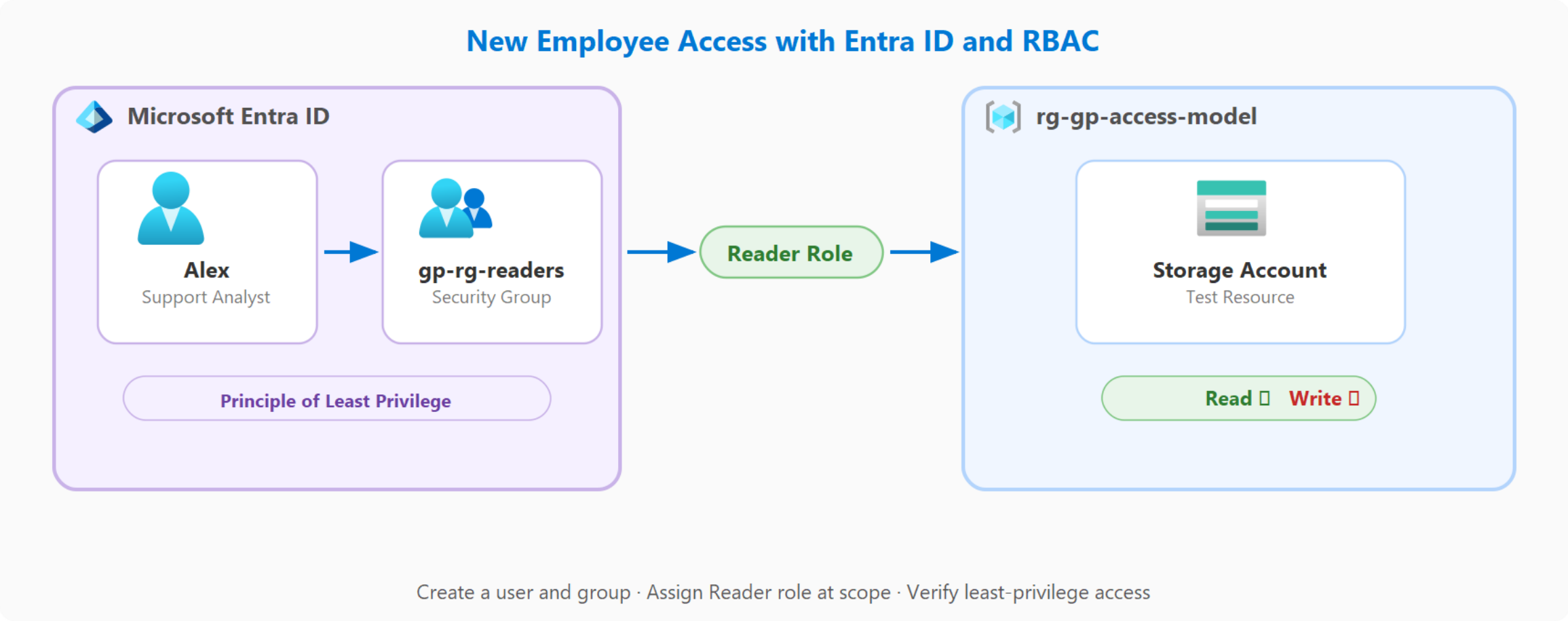
After completing this module, you will be able to:

- 1** Create a user and security group in Microsoft Entra ID
- 2** Assign group-based Azure RBAC at resource group scope
- 3** Validate expected access at the correct scope

Project: Identity and RBAC onboarding objectives

- Scenario: A new support analyst needs read-only access to a development resource group
- Microsoft Entra ID manages users and groups; RBAC assigns specific permissions at a scope
- Together they implement the principle of least privilege
- Exercise 1: Create a user account and security group in Entra ID
- Exercise 2: Assign the Reader role to the group at resource group scope
- Exercise 3: Verify read access is granted and write access is denied

Project: Identity and RBAC onboarding diagram



Knowledge check



1 What is the benefit of assigning RBAC roles to a group instead of individual users?

- ☐ Groups automatically inherit subscription permissions
- ☐ It simplifies access management when users join or leave
- ☐ Groups can hold more permissions than individual users

2 What RBAC role provides view-only access to Azure resources?

- ☐ Contributor
- ☐ Owner
- ☐ Reader

3 At what scope level was the RBAC role assigned in this guided project?

- ☐ Subscription
- ☐ Resource group
- ☐ Individual resource

Guided project: Share files securely



Learning Objectives

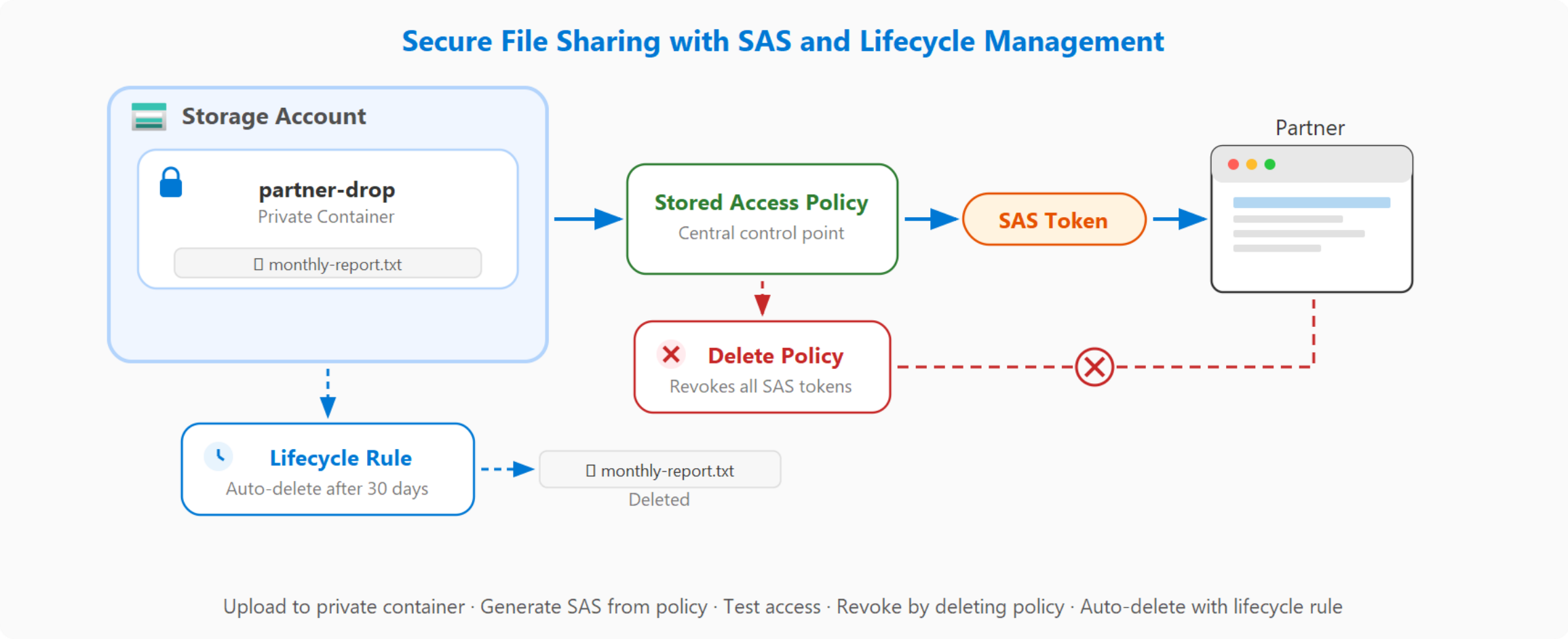
After completing this module, you will be able to:

- 1** Create blob storage and generate policy-based SAS links for secure sharing
- 2** Validate SAS access and revoke it instantly by deleting a stored access policy
- 3** Configure lifecycle management rules to auto-delete shared files

Project: Secure file sharing with SAS objectives

- Scenario: Share monthly reports with an external partner for a short review window
- Stored access policies let you centrally manage and revoke SAS tokens instantly
- Exercise 1: Create a storage account, private container, and upload a file
- Exercise 2: Create a stored access policy and generate a SAS link
- Exercise 3: Test SAS access and verify direct access is blocked
- Exercise 4: Revoke access by deleting the stored access policy
- Exercise 5: Configure lifecycle management for automatic cleanup

Project: Secure file sharing with SAS diagram



Knowledge check



1 How can you instantly revoke all SAS tokens generated from a stored access policy?

- ☐ Regenerate the storage account access keys
- ☐ Delete the stored access policy
- ☐ Set the SAS token expiry to a past date

2 What determines the permissions and expiry of a policy-based SAS token?

- ☐ The storage account access tier
- ☐ The stored access policy
- ☐ The container public access level

3 What does a lifecycle management rule do for blob storage?

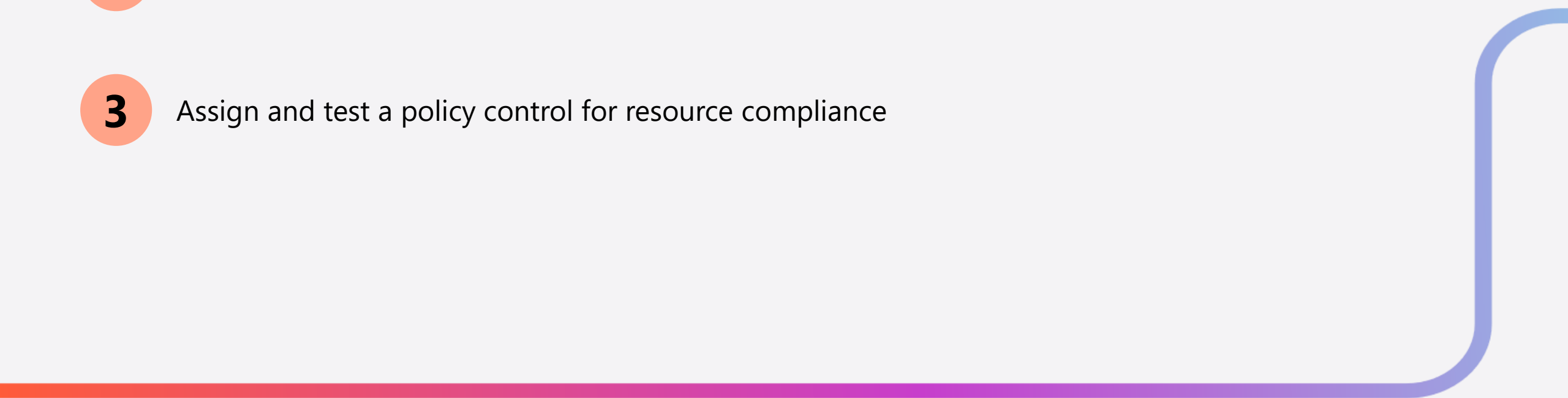
- ☐ Moves blobs between storage accounts
- ☐ Backs up blobs to a secondary region
- ☐ Automatically deletes or moves blobs based on age

Guided project: Set up cost guardrails in Azure



Learning Objectives

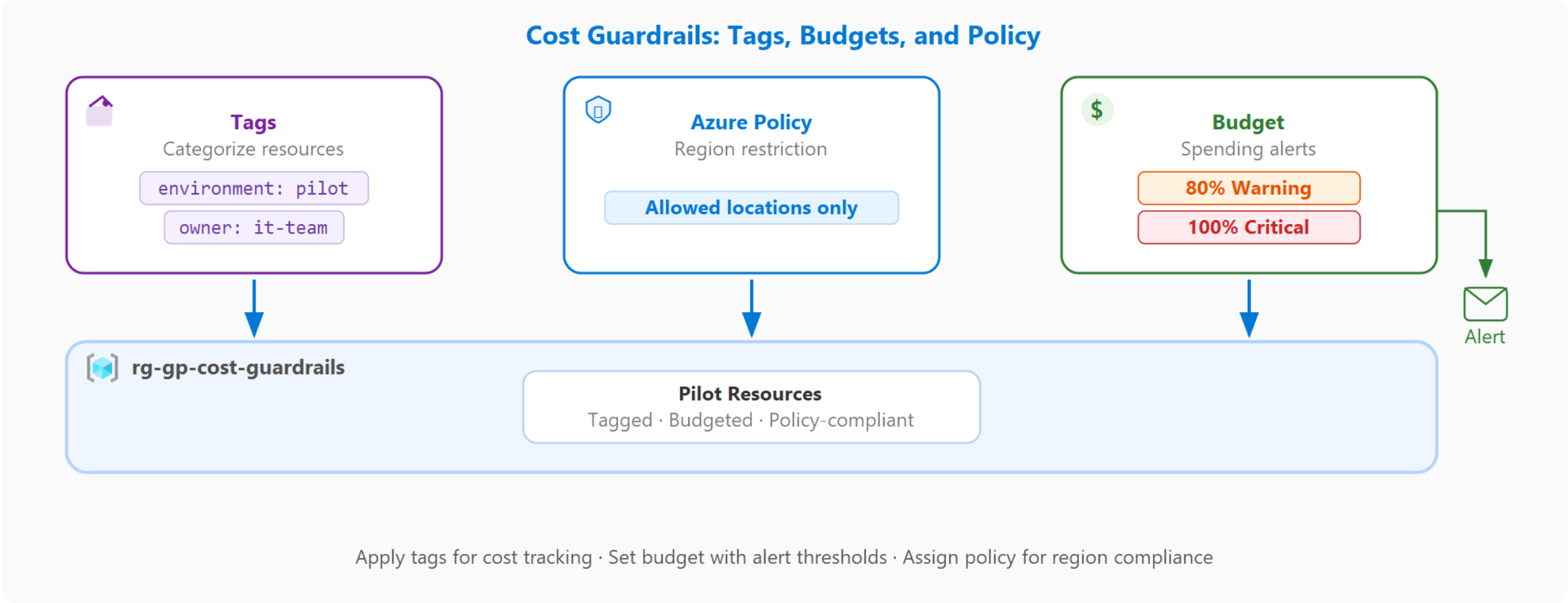
After completing this module, you will be able to:

- 1** Apply consistent resource tags for cost tracking
 - 2** Create budget alerts for early spend visibility
 - 3** Assign and test a policy control for resource compliance
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Project: Cost guardrails with tags, budgets, and policy objectives

- Scenario: A pilot team needs clear cost controls before a broader cloud rollout
- Tags categorize resources for cost reporting; budgets alert when spending reaches thresholds
- Azure Policy enforces organizational rules like restricting deployment regions
- Exercise 1: Apply organizational tags for cost tracking
- Exercise 2: Create a budget with alert thresholds
- Exercise 3: Assign and test an Azure Policy for region compliance

Project: Cost guardrails with tags, budgets, and policy diagram



Knowledge check



1 What happens when a resource deployment violates an Azure Policy with a deny effect?

- ☐ The deployment succeeds but is flagged as noncompliant
- ☐ The deployment is blocked
- ☐ The resource is deployed and then automatically deleted

2 What is the purpose of a budget alert in Azure?

- ☐ To automatically stop all running resources
- ☐ To notify you when spending approaches a defined threshold
- ☐ To enforce a hard spending cap

3 Why should you apply consistent tags to Azure resources?

- ☐ Tags are required before you can deploy resources
- ☐ Tags automatically reduce the cost of resources
- ☐ To track and allocate costs by team, environment, or project

Guided project: Monitor Azure with Service Health and Activity Log alerts



Learning Objectives

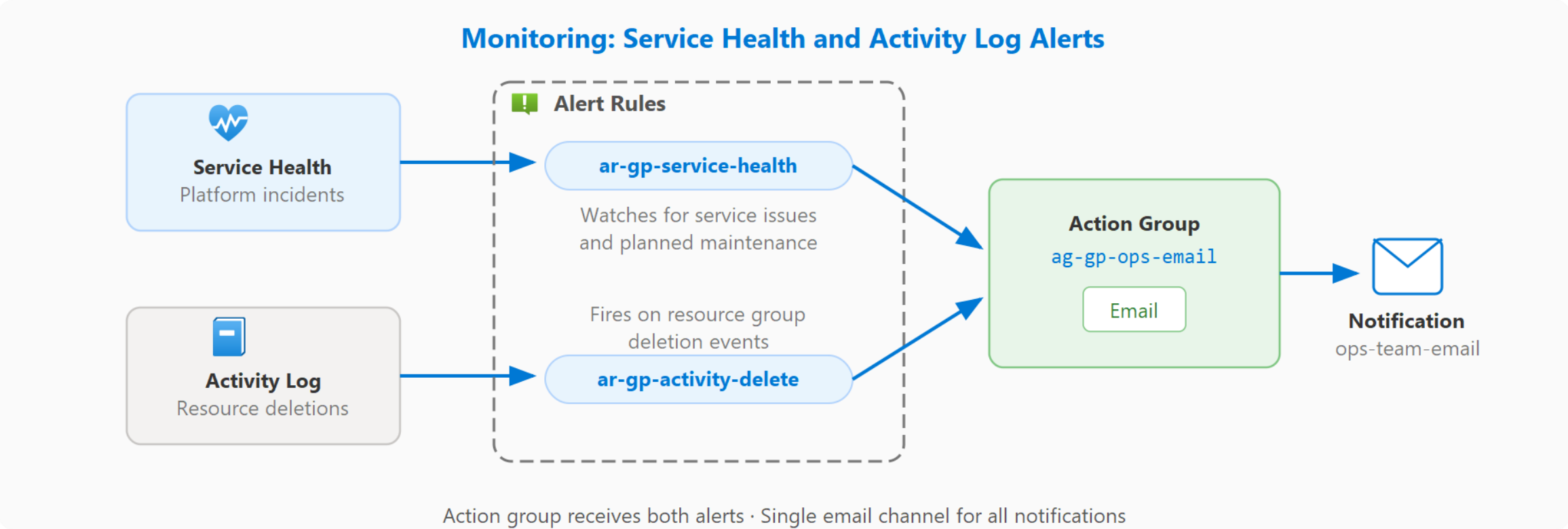
After completing this module, you will be able to:

- 1** Create an action group with email notifications
- 2** Create a Service Health alert for platform incidents and maintenance
- 3** Create an Activity Log alert for resource deletion events

Project: Monitoring with alerts and action groups objectives

- Scenario: Notify the operations team when Azure platform issues occur or resources are deleted
- Action groups define who gets notified and how when an alert fires
- Service Health alerts watch for platform-level incidents and planned maintenance
- Activity Log alerts fire when management operations like resource deletion occur
- Exercise 1: Create an action group and verify email delivery
- Exercise 2: Create a Service Health alert for platform incidents
- Exercise 3: Create an Activity Log alert for resource deletions

Project: Monitoring with alerts and action groups diagram



Knowledge check



- 1** What is the purpose of an action group in Azure Monitor?
 - ☐ To define which metrics are collected
 - ☐ To define who gets notified and how when an alert fires
 - ☐ To group related alert rules into a dashboard

- 2** What type of alert monitors Azure platform incidents and planned maintenance?
 - ☐ Activity Log alert
 - ☐ Metric alert
 - ☐ Service Health alert

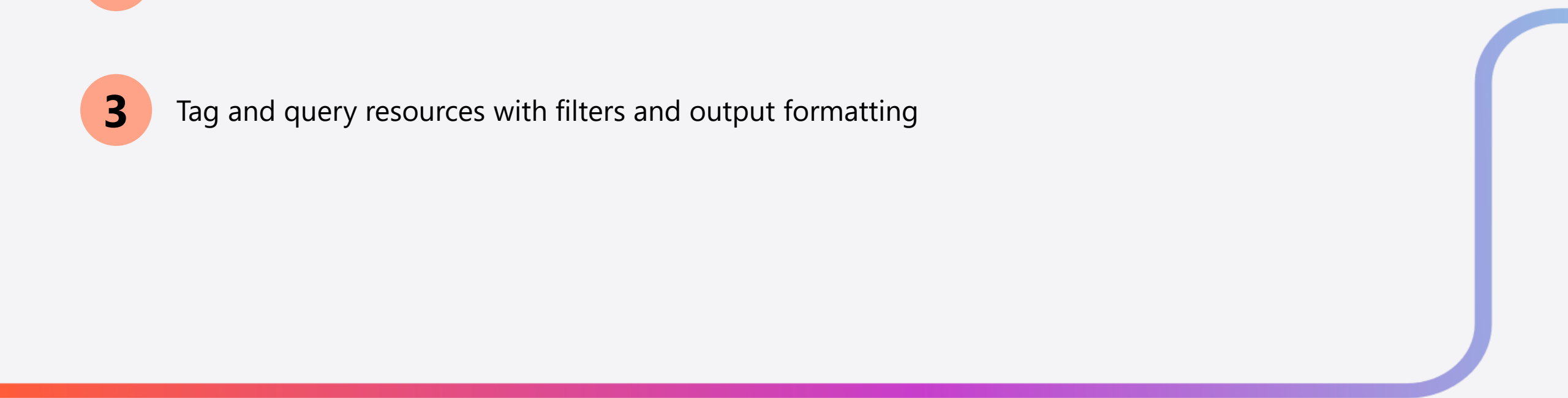
- 3** What type of event does an Activity Log alert detect?
 - ☐ Changes in resource performance metrics
 - ☐ Management operations such as resource creation or deletion
 - ☐ User sign-in attempts to the Azure portal

Guided project: Manage Azure resources with Cloud Shell and the Azure CLI



Learning Objectives

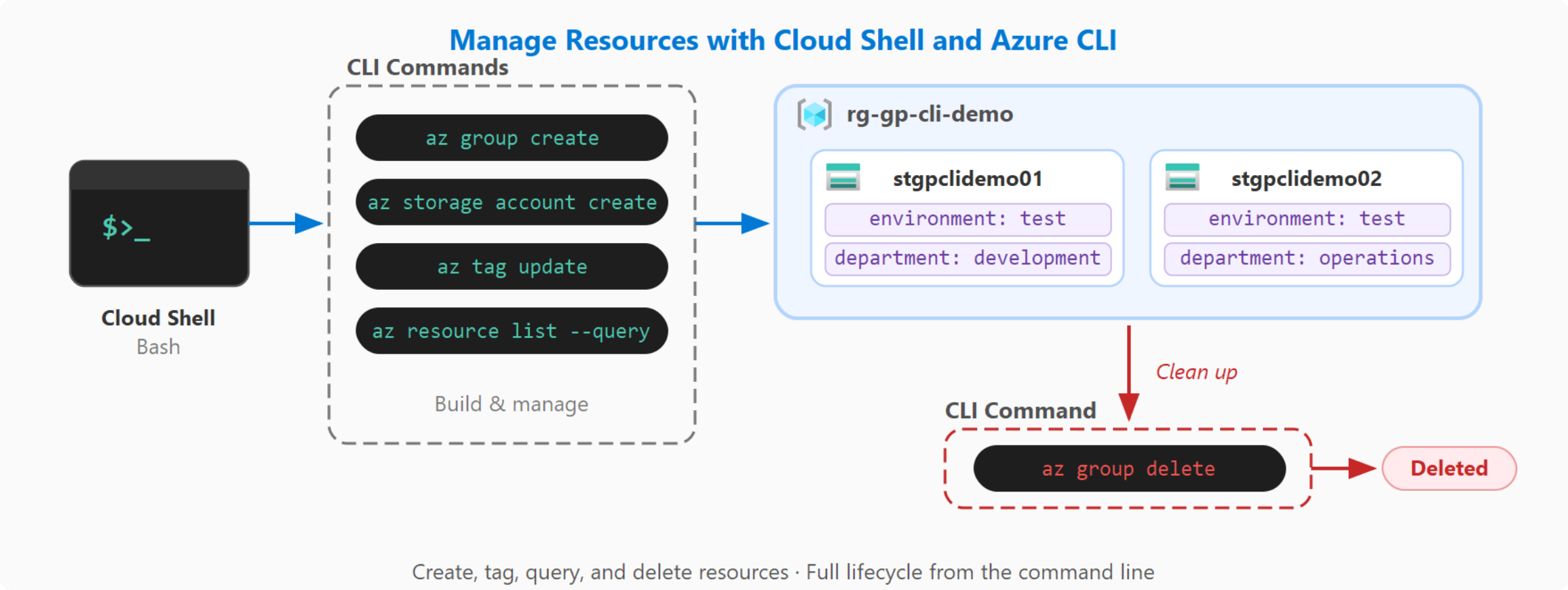
After completing this module, you will be able to:

- 1** Open Cloud Shell and navigate the Bash environment
 - 2** Create and list Azure resources using CLI commands
 - 3** Tag and query resources with filters and output formatting
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Project: Resource management with Cloud Shell and CLI objectives

- Scenario: Manage resources efficiently from the command line instead of navigating the portal
- Cloud Shell is a browser-based terminal with the Azure CLI preinstalled
- The CLI is useful when performing the same operations across multiple resources quickly
- Exercise 1: Open Cloud Shell and explore the environment
- Exercise 2: Create and list resources with CLI commands
- Exercise 3: Tag, query, and clean up resources from the command line

Project: Resource management with Cloud Shell and CLI diagram



Knowledge check



- 1** What query language does the Azure CLI use to filter and shape JSON output?
 - ☐ SQL
 - ☐ JMESPath
 - ☐ KQL (Kusto Query Language)

- 2** What happens when you delete a resource group using the Azure CLI?
 - ☐ Only the resource group is deleted
 - ☐ All resources inside the resource group are also deleted
 - ☐ Resources are moved to a default resource group

- 3** What shell environment does Cloud Shell use by default for Azure CLI commands?
 - ☐ PowerShell
 - ☐ Command Prompt
 - ☐ Bash